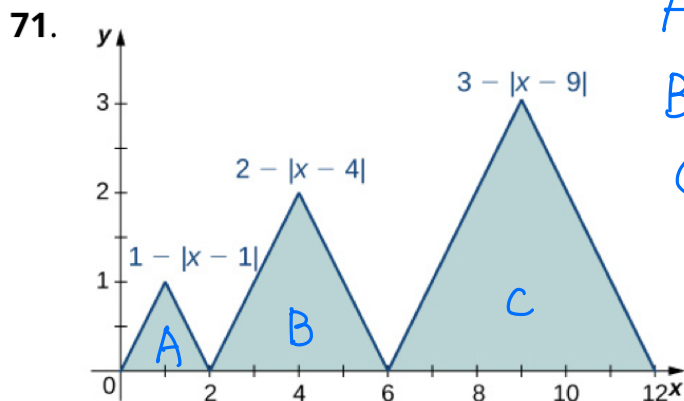


In the following exercises, evaluate the integrals of the functions graphed using the formulas for areas of triangles and circles, and subtracting the areas below the x-axis.

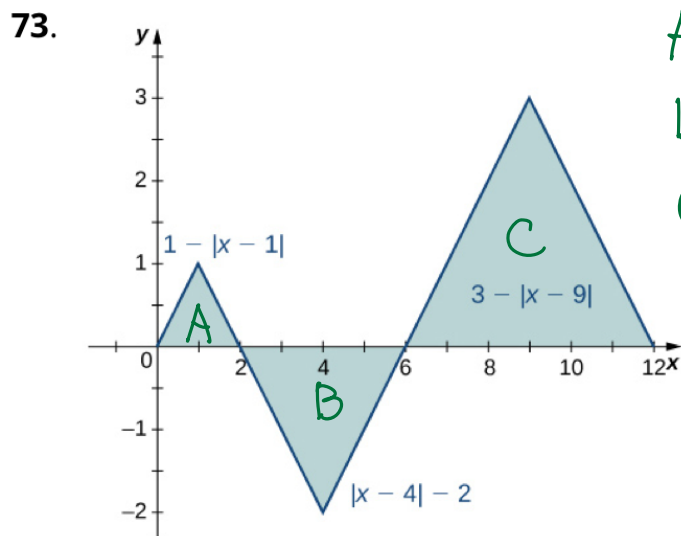


$$A = 2 \times 1 \times \frac{1}{2} = 1$$

$$B = (6-2) \cdot 2 \cdot \frac{1}{2} = 4$$

$$C = (12-6) \cdot 3 \cdot \frac{1}{2} = 9$$

$$A+B+C = 1+4+9 = 14$$

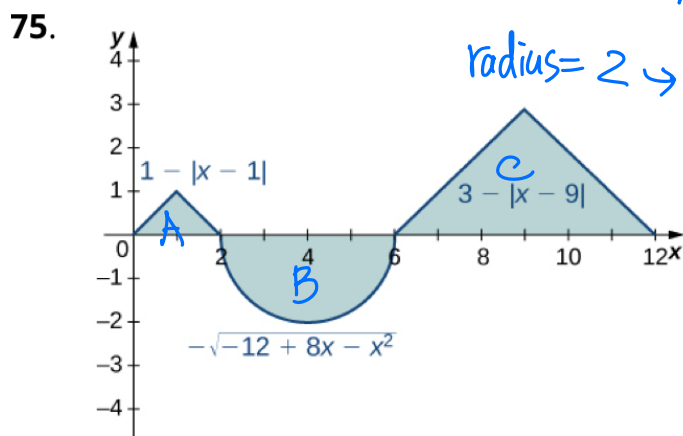


$$A = 2 \times 1 \times \frac{1}{2}$$

$$B = (6-2) \times 2 \times \frac{1}{2} = 4$$

$$C = (12-6) \times 3 \times \frac{1}{2} = 9$$

$$\text{Shaded Area} = A - B + C = 6$$



$$A = 2 \times 1 \times \frac{1}{2} = 1$$

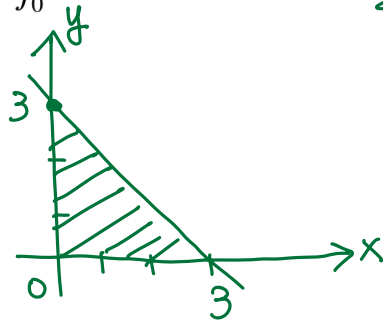
$$\text{radius} = 2 \rightarrow B = 2^2 \cdot \pi \cdot \frac{1}{2} = 2\pi$$

$$C = (12-6) \cdot 3 \cdot \frac{1}{2} = 9$$

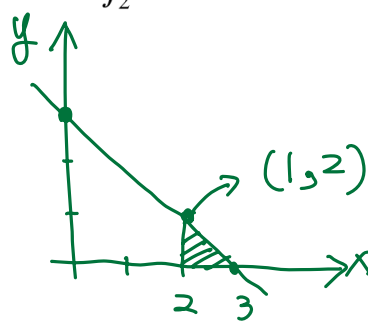
$$\text{Shaded Area} = A - B + C = 10 - 2\pi$$

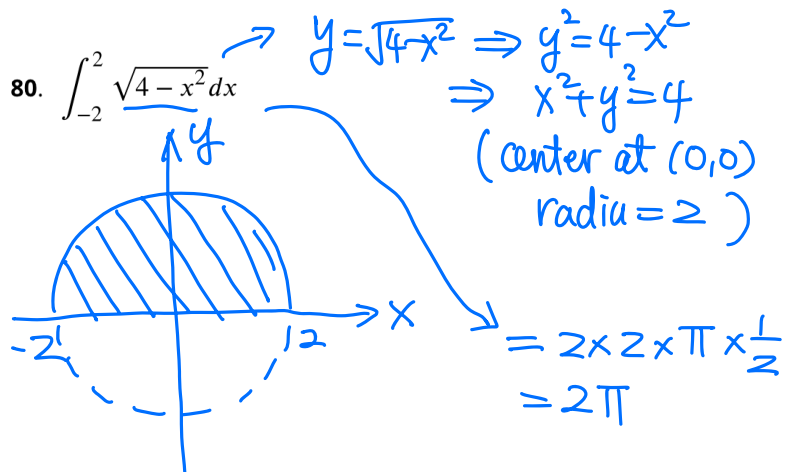
In the following exercises, evaluate the integral using area formulas.

76. $\int_0^3 (3-x) dx = 3 \times 3 \times \frac{1}{2} = \frac{9}{2}$



77. $\int_2^3 (3-x) dx = 1 \times 1 \times \frac{1}{2} = \frac{1}{2}$





Suppose that $\int_0^4 f(x) dx = 5$ and $\int_0^2 f(x) dx = -3$, and $\int_0^4 g(x) dx = -1$ and $\int_0^2 g(x) dx = 2$. In the following exercises, compute the integrals.

88. $\int_0^4 (f(x) + g(x)) dx$

$$= \int_0^4 f(x) dx + \int_0^4 g(x) dx$$

$$= 5 + (-1)$$

$$= 4$$

89. $\int_2^4 (f(x) + g(x)) dx$

$$= \int_2^4 f(x) dx + \int_2^4 g(x) dx$$

$$= \int_0^4 f(x) dx - \int_0^2 f(x) dx$$

$$+ \int_0^4 g(x) dx - \int_0^2 g(x) dx$$

$$= 5 - (-3) + (-1) - (2)$$

$$= 8 - 1 - 2 = 5$$

90. $\int_0^2 (f(x) - g(x)) dx$

$$= \int_0^2 f(x) dx - \int_0^2 g(x) dx$$

$$= -3 - (2)$$

$$= -5$$

92. $\int_0^2 (3f(x) - 4g(x)) dx$

$$= \int_0^2 3f(x) dx - \int_0^2 4g(x) dx$$

$$= 3 \int_0^2 f(x) dx - 4 \int_0^2 g(x) dx$$

$$= 3(-3) - 4(2)$$

$$= -9 - 8 = -17$$